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Diffusion maps based k-nearest-neighbor rule technique for semiconductor manufacturing process fault detection

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ABSTRACT

In the semiconductor industry, traditional multivariate statistical process monitoring methods and pattern classification based detection methods have been developed to detect the semiconductor process faults. However, they do not show superior performance due to the limits of these methods and the unique characteristics of semiconductor processes such as non-linearity and multimodal batch trajectories. This paper presents a novel diffusion maps based k-nearest-neighbor rule (DM-kNN) technique that can reduce data-storage costs and enhance the performance of fault detection by integrating diffusion maps analysis with k-nearest-neighbor rule. DM-kNN takes full advantage of the dimensionality reduction and information preserving properties of DM to extract the low dimensional manifold feature that optimally preserves the intrinsic nonlinear structure of the data set. Then the adapted kNN rule based fault detection method is applied to the low dimensional manifold feature space to detect potential faults. The effectiveness and robustness of DM for dimensionality reduction and feature extraction are verified in simulation experiments compared with other linear and nonlinear dimensionality reduction methods. In addition, DM-kNN is applied to monitor the semiconductor manufacturing process. The fault detection results of the proposed method are demonstrated to be superior to those of the MPCA, FD-kNN, PC-kNN and FS-kNN approaches.

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1. Introduction

Semiconductor manufacturing system is considered to be one of the most complicated high-tech manufacturing systems. The key characteristics of this process include the high complexity of wafer processing, non-linearity of most batch processes, multimodal batch trajectories due to product mix, large-scale systems and the high degree of measurement uncertainty, which all together have posed great difficulties to these traditional fault detection methods [1]. Therefore, designing effective fault detection technologies to improve product quality, reduce scrap rate and wafer quantity, and ensure the safety during the production process is the key point of the semiconductor manufacturing process. Many advanced process controls and detecting methods are proposed to obtain accurate and effective fault detection results [2–5].

Data-driven based statistical process control methods have been widely applied in semiconductor manufacturing processes, because of large amounts of trace or machine data generated and collected by process equipments in modern industry [6]. Among all fault detection methods, multivariate statistical process control (MSPC) method as the effective data-driven method, has been successfully applied to on-line process fault detection, in particular, chemical processes, biochemical processes, semiconductor processes, etc. [7–9]. The most popular MSPC methods are multi-way PCA (MPCA) and multi-way PLS (MPLS) in the semiconductor industry [6,10–13]. However, some drawbacks of MPCA and some unique characteristics of semiconductor manufacturing processes make it unable to gain ideal detection performance. The main reason is that the derivation of threshold for the MPCA detection statistics relies on the assumption that the process data follow a multivariate Gaussian distribution approximately [14,15]. Nevertheless, the unique characteristics of semiconductor manufacturing processes such as non-linearity and multimodal batch trajectories usually result in non-Gaussian distributed process information, which will deteriorate the detection performance of MPCA and MPLS. To overcome this limitation and explicitly account for these unique characteristics, a pattern recognition based fault detection method utilizing the k-nearest-neighbor rule (FD-kNN) has been developed recently [16]. The basic idea of FD-kNN is that a faulty sample's distance to the k nearest neighboring normal samples must be greater than a normal sample's distance to the k nearest neighboring normal samples. FD-kNN can offer better fault detection performance compared with MPCA. However, the drawback of FD-kNN is that it requires large storage space to store the complete original training data set for large-scale processes with thousands of variables after batch unfolding. This is because for the test sample, FD-kNN needs to search all the training data set to calculate the Euclidean distance between test sample and training data set to further identification of its kNN. This may increase the burden of on-line process fault detection when tens of thousands

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of such fault detection models are running simultaneously due to high-mix production. For example, IBM has more than 7000 on-line fault detection models of running [17], while Intel has as many as 30,000 of such models [18]. In order to reduce storage space, He and Wang [19] proposed principal component based kNN method (PC-kNN). The basic idea was using PCA to realize data dimension reduction and extract the original samples' principal component subspace (PCs) which was the modeling sample, then the FD-kNN method is applied to the score subspace to detect faults. However, this method ignores abnormal information that occurred in the residual subspace, leading to high Type II errors (missed detection). Aiming at the flaws that the PC-kNN ignores the residual information, feature space based k nearest neighbor method (FS-kNN) was proposed by Guo et al. [20]. In their work, the feature space was the sum of PC subspace and squared prediction error (SPE). Compared to PC-kNN, FS-kNN has an improved fault detection performance, but there is still a high false alarm and missed detection rates. The main reason of suboptimum fault detection performance showed by FS-kNN and PC-kNN is that the principal component subspace or feature space extracted by PCA may not be able to effectively represent the original data space. On the one hand, PCA is a kind of linear technique; it may not be able to extract nonlinear relations or features of the complex semiconductor data with severe non-linearity. On the other hand, PCA is a second-order method, which means that it considers only mean and variance or covariance of the data set, but does not take into account the local structure information from the data with complicated distributions, which is common for the actual industrial processes.

To handle the non-linear problem of industrial process data, some nonlinear theories have been developed, such as the conventional nonlinear PCA, neural networks and other kernel-based approaches. Dong and McAvoy [21] developed a nonlinear PCA combined with principal curves and neural networks. However, such nonlinear mappings can only be suitable for a limited class of nonlinear models due to the nonlinear function of the principal curve algorithm which is a linear combination of some univariate functions [22]. Furthermore, a nonlinear optimization problem cannot be avoided to calculate the principal curves and train the neural networks. Recently, the kernel PCA has been proposed through introducing the nonlinear kernel function, which does not involve nonlinear optimization [23–26]. However, the possibility that the intrinsic geometry structure of data may reside on a manifold is not explicitly considered by KPCA [27]. Recently, some promising nonlinear dimensionality reduction algorithms called manifold learning have emerged, such as local tangent space alignment (LTSA) [28], locally linear embedding (LLE) [29], isometric feature mapping (ISOMAP) [30], and diffusion maps (DM) [31,32]. Among them, diffusion maps approach is a robust nonlinear manifold dimensionality reduction technique. It is defined on a graph of the data points through constructing a Markov chain. Then, the spectral eigenanalysis is performed on the probability transition matrix of the Markov chain to achieve the low-dimensional representation of data points. Compared with the classical linear technique, DM can explicitly discover the nonlinear manifold geometry that embedded in the complex high-dimensional space. Simultaneously, it preserves the local structure of the features via retaining the pairwise diffusion distances as well as possible. Moreover, diffusion distance is computed using all possible paths between any two data points, so it is more robust to noise than the geodesic distance (shortest path distance) used by ISOMAP. At present, the application fields of diffusion maps are mostly focused on image processing and pattern recognition. For instance, Jingen Liu et al. [33] applied diffusion maps analysis to learn the semantic features for action recognition; Feng Zheng et al. [34] proposed a semi-supervised algorithm based on diffusion maps to realize visualization on the Yale face and human pose images; and R. R. Coifman et al. [35] presented diffusion maps technique to identify different tissue samples of hyperspectral pathology tissue image. It noticed that the target dimensionality of above applications is always restricted not to go beyond three in order to realize visualization, this may lead to poor performance due to excessive discard of some meaningful information. Hence, it is necessary to estimate the intrinsic dimensionality of the original data during the feature extraction phase. Furthermore, diffusion maps technique is seldom researched in the fault diagnosis field, especially the application in the industrial process fault detection and diagnosis is much less. Yixiang Huang et al. [36] used discriminant diffusion maps analysis for machinery condition monitoring and fault diagnosis.

In order to enhance the performance of fault detection, reduce the storage of the training samples and overcome the shortcomings of PCA in dealing with non-linear data sets, a novel fault detection method called diffusion maps based k-nearest-neighbor rule (DM-kNN) is presented for semiconductor manufacturing process in this paper. In the first step, correlation dimension method is used to estimate intrinsic dimensionality of the original data set, and then diffusion maps analysis is applied to the original data set to transform high dimensional data into low dimensional manifold feature space with the intrinsic dimensionality. In the second step, the kNN fault detection method is applied to low dimensional manifold feature space to detect potential faults. From the viewpoint of feature extraction and fault detection performance, DM-kNN has the following advantages:

1) Diffusion maps analysis is capable of discovering the nonlinear structure of the manifold existing in the given data and optimally preserves the intrinsic geometry structure in a low dimensional manifold feature space.
2) DM-kNN has relatively high robustness and stability for the process fault detect, because the diffusion maps algorithm based on the diffusion distance has more robust to noise than, e.g., the geodesic distance.
3) Normal and abnormal samples can be obviously mapped into different manifold regions after diffusion maps and kNN detection statistics is based on distance, which causes DM-kNN to have an optimal fault detection performance.
4) Compared to FD-kNN, DM-kNN can remove the redundancy of the original data sets and reduce storage space since the low manifold feature space usually has a much lower dimension compared with the original data space.

The rest of this paper is organized as follows. Section 2 reviews the FD-kNN technique. Section 3 describes the theoretical aspects of diffusion maps analysis and presents the novel fault detection method of DM-kNN. In Section 4, simulation example is, firstly, used to illustrate dimensionality reduction and information extraction capability of the DM method. Then, DM-kNN is applied to semiconductor manufacturing process, and the fault detection results are compared with MPCA, FD-kNN, PC-kNN and FS-kNN approaches. The conclusions of this work are summarized in Section 5.

2. Fault detection using k-nearest-neighbor rule (FD-kNN)

The k-nearest neighbor (kNN) rule was first developed by Fix and Hodges [37]. kNN has been widely used in pattern classification such as near infrared (NIR) spectroscopy data classification [38,39], where the unlabeled samples are classified through searching the k nearest labeled samples in the training data. Recently, a new fault detection method using kNN rule (FD-kNN) has been developed. The basic idea of FD-kNN lies in the fact that a normal sample trajectory is similar to the trajectories of normal training data. There must be some deviation between the trajectory of a fault sample and a normal one in the training data. In other words, a fault sample's distance to the nearest neighboring training samples must be greater than that of a normal sample. This detection strategy is implemented by calculating the average squared distance from a sample to its k nearest neighbors in the normal training samples. If a threshold with the certain confidence level is determined based on the distribution of normal training samples'

distances to their k nearest neighboring training samples, this threshold can be utilized to detect the unclassified sample. If its distance to its nearest neighboring training samples is below the threshold, the sample is considered as normal. Otherwise, it is a fault.

2.1. Model building

For a training data set $\mathbf{X}\{\infty x_1 x_2 \cdots x_N\} \in R^{N \times d}$, $(i, j = 1, 2, \cdots N)$, where N is the number of samples and d as the original dimension.

1. Find k nearest neighbors for each sample in the training data set.
2. Calculate the kNN squared distance of each sample x_i as defined by

$$D_i^2 = \frac{1}{k}\sum_{j=1}^{k} d_{ij}^2 \tag{1}$$

   where D_i^2 is the sum of squared distances of the sample x_i to its k-nearest neighbors and d_{ij}^2 denotes squared Euclidean distance from the sample x_i to its jth nearest neighbor.
3. Determine the corresponding control limit of D_i^2 for fault detection.

The threshold D_α^2 with a significance level α can be determined by an assumption that D_i^2 following a non-central χ^2 distribution [16].

2.2. Fault detection

For a new sample x to be detected,

1. Identify x's k nearest neighbors in the training data set.
2. Calculate x's kNN squared distance D_x^2.
3. Compare D_x^2 with the threshold D_α^2, if $D_x^2 \le D_\alpha^2$, it is considered as a normal sample. Otherwise, it is detected as a fault.

More detailed discussion on the FD-kNN method can be found in [16].

3. Diffusion maps based k-nearest-neighbor rule approach for fault detection

The disadvantage of FD-kNN is that it requires larger storage space to store the training data sets for large-scale processes, because there may be thousands of variables after batch unfolding. The reason for large requirement of storage space is that for each test sample, FD-kNN requires finding the sample's k nearest neighbors from the whole training data set through computing the Euclidean distance between test sample and training data set. Although PC-kNN was proposed in order to reduce the storage space, it could not show the excellent performance. On the one hand, PC-kNN uses the scores of all training data in the PCs to represent the original normal training processes, which can only capture the linear behavior of the process operation. On the other hand, PC-kNN makes no use of the residual information, thus any fault that occurs in the residual subspace will not be detected. Compared with PC-kNN, although FS-kNN takes the residual information into account, it still shows a relatively low rate of fault detection.

In this study, diffusion maps based k-nearest-neighbor rule (DM-kNN) approach is proposed to detect semiconductor manufacturing process faults by integrating diffusion maps analysis with k-nearest-neighbor rule. The proposed method can remove the redundancy of the original data set and reduce the storage space of the training samples compared with FD-kNN. In contrast to MPCA, PC-kNN and FS-kNN, DM-kNN does not require any assumptions regarding the measurement variables and can well deal with the process data with nonlinear characteristics. In addition, the low dimensionality manifold feature space with intrinsic dimensionality is extracted by diffusion maps can effectively represent the original high-dimensional data space. In this way, DM-kNN shows perfect and robust fault detection capability.

3.1. Diffusion maps analysis

The diffusion maps framework derives from the field of dynamical systems [40,41]. In the first step, it needs to establish a neighbor graph according to the high dimensional spatial data. Then a Markov random walk on the graph of data is defined. By performing certain time step random walk; a metric represents the proximity of the data points is obtained. This metric is defined as the diffusion distance. The low dimensional embedding manifold of the data could be obtained through maximizing retain the diffusion distance.

For the data set $\mathbf{X}$, a weighted symmetrical graph $\mathbf{G}(\mathbf{X}, \mathbf{W})$ needs to be constructed, in which each x_i corresponds to a node, and $\mathbf{W} = \{w_{ij}(x_i, x_j)\}$ is its weighted adjacency matrix which is symmetrical and positive. Each entry $w_{ij}(x_i,x_j)$ of $\mathbf{W}$ represents the degree of similarity or affinity of two data points and can be computed using a weight function $k(x_i,x_j)$. The weight function should satisfy the conditions of symmetry, positivity and positive semi-definite; that is $k(x_i,x_j) = k(xj,xi)$, $k(x_i,x_j) \ge 0$ and $\int_x \int_x k(x_i, x_j) g(x_i) g(x_j) du(x_i) du(x_j) \ge 0$, respectively. In this work, an isotropic Gaussian diffusion kernel is chosen, leading to the weight matrix with entries:

$$w_{ij} = \exp\left(-\frac{\left\|x_i - x_j\right\|^2}{2\sigma^2}\right) \tag{2}$$

where σ is the bandwidth of the Gaussian kernel. Under an equal kernel width, it is clear that the farther the distance between the two data points, the smaller the weight. At the same time, the kernel width controls the degree of diffusion, the larger the kernel width, the more neighbors of a data point. Hence, the matrix $\mathbf{W}$ represents the information of the local geometry of the data set.

Subsequently, the edge weight matrix $\mathbf{W}$ should be normalized, and its rows are added up to 1. In this way, a matrix $\mathbf{P} = \mathbf{P}^{(1)} = \{p_{ij}^{(1)}\}$ is constructed with entries:

$$p_{ij}^{(1)} = \frac{w_{ij}}{\sum_k w_{ik}.} \tag{3}$$

Since diffusion maps originate from dynamical system theory, the matrix $\mathbf{P}$ can be interpreted as defining a random walk on the graph $\mathbf{G}$, which governs the evolution of the chain on the space $\mathbf{X}$. Therefore, $\mathbf{P}^{(1)}$ represents the transition probability from x_i to x_j in a single time step, and $\mathbf{P}$ represents the entire Markov chain. The forward probability matrix for t time steps, i.e., $\mathbf{P}^{(t)}$, is given by $[\mathbf{P}^{(1)}]^t$. The increment of t can be regarded as the propagation of the local influence of each node on its neighbors [33].

Then the diffusion distance $\mathbf{D}$ between two data points on the graph $\mathbf{G}$ can be defined by the random walk forward probabilities $p_{ij}^{(t)}$ as follows:

$$\mathbf{D}^{(\mathbf{t})}\left(x_i, x_j\right) = \sqrt{\sum_{k \in X} \frac{\left(p_{ik}^{(t)} - p_{jk}^{(t)}\right)}{\varphi(x_k)^{(0)}}}, \varphi(x_k) = \frac{m_k}{\sum_j m_j}, m_k = \sum_j p_{kj} \tag{4}$$

where $\varphi(x_k)$ is the unique stationary distribution which measures the density of the data features [18] and m_k is the degree of a node x_k. From Eq. (4), it can be noted that pairs of data points with a high forward transition probability have a small diffusion distance. In other words, diffusion distance yields information regarding how many paths exist between points x_i and x_j, so the value of diffusion distance will be small if the points are connected by many paths in the graph. For instance, assume that there are two-dimensional data points (see

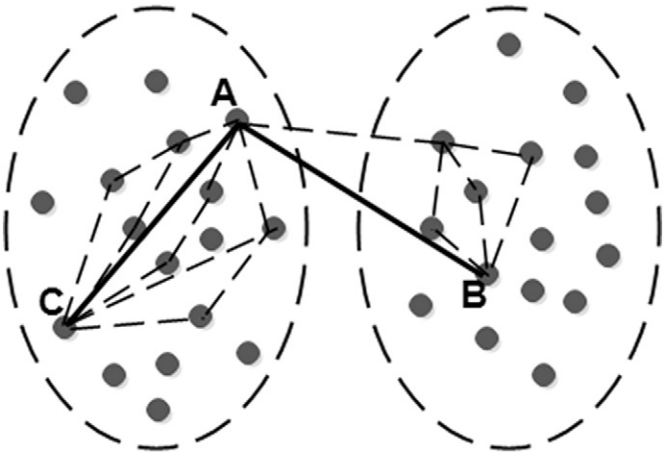


Fig. 1. The difference between diffusion distance and Euclidean distance.

Fig. 1). The distance from points A to B equals to that between A and C in terms of Euclidean distance, while the diffusion distance between A and C is shorter than that from A to B. This is mainly because the possible random paths from A to C are much more than those from A to B, which leads to the small value of the diffusion distances. This example illustrates that the diffusion distance can reflect the local geometric structure of data. Another notable point of the diffusion distance is that it is computed by all possible paths through the graph; therefore, the contribution of all the edges connecting the two points is taken into account. This makes the diffusion distance much more robust to noise than, e.g., the geodesic distance.

3.2. Dimensionality reduction using the diffusion maps

Dimensionality reduction is important during the feature extraction phase of high-dimensional data for process fault detection. It can avert the curse of dimensionality and reduce the redundancy of high dimensional spaces. Diffusion maps technique has been proposed to deal with the nonlinear process for dimensionality reduction. It aims at identifying and extracting the low-dimensionality embedding manifold which the data lies on. Ideally, the low-dimensionality representation has a dimensionality that equals to the intrinsic dimensionality of the data. Intrinsic dimensionality is the minimum number of parameters that is needed to account for the meaningful properties of the data. The methods that have been proposed for intrinsic dimensionality estimation up to now can be subdivided into two types. The first one is based on the dominant eigenvalues. The other type is the fractal-based method, which is initially used to estimate the dimension of a chaotic dynamical system. It is generally characterized by a non-integer dimensionality [36].

The dominant-eigenvalues method uses the first l nontrivial eigenvalues with a relative precision δ to determine the intrinsic dimensionality.

$$d' = \max\left\{l | l \in N, \lambda_l^t > \delta\lambda_1^t\right\} \tag{5}$$

However, the main drawback of the dominant-eigenvalues method is that one has to define a threshold for the dominant eigenvalues to make the sum-of-squares error function minimum. This method trends to overestimate the intrinsic dimensionality and cannot offer reliable results [36]. Therefore, the correlation dimension method which belongs to a type of the fractal-based methods is implemented to determine the intrinsic dimension [42].

The correlation dimension estimator is based on the observation that the number of data points in a hyper-sphere around a data point with radius r grows proportional to $r^{d'}$, where d' is the intrinsic dimensionality of the data manifold around that data point. The intrinsic dimensionality d' can be obtained by computing the relative amount of data points that lie within a hyper-sphere with radius r. The relative amount of data points that in a hyper-sphere with radius r is defined by

$$C(r) = \frac{2}{n(n-1)}\sum_{i=1}^{n}\sum_{j=i+1}^{n} c \quad \text{where} \quad c = \begin{cases} 1, if \left\|x_i - x_j\right\| \le r \\ 0, if \left\|x_i - x_j\right\| > r \end{cases}. \tag{6}$$

Because $C(r)$ is proportional to $r^{d'}$, we can use $C(r)$ to estimate the intrinsic dimensionality d' of the data. The intrinsic dimensionality d' is defined by the limit

$$d' = \lim_{r\to 0}\frac{\log C(r)}{\log r}. \tag{7}$$

In order to avoid a solution of limit and considering the nature of the logarithmic function, the intrinsic dimensionality d' can be estimated by computing $C(r)$ for two different values of r.

$$d' = \frac{\log(C(r_2) - C(r_1))}{\log(r_2 - r_1)} \tag{8}$$

After the intrinsic dimensionality d' is determined, the d' dimensionality nonlinear manifold feature space can be obtained by performing diffusion maps. In the process of embedding, diffusion maps attempt to retain the diffusion distances.

Using the spectral eigen-decomposition on the transition matrix $\mathbf{P}$, we are able to obtain a series of left eigenvectors and right eigenvectors.

$$\begin{cases} \mathbf{L}_n^T\mathbf{P}^{(t)} = \lambda_n\mathbf{L}_n^T \\ \mathbf{P}^{(t)}\mathbf{R}_n = \lambda_n\mathbf{R}_n \end{cases}, \tag{9}$$

Where $\mathbf{L}_n$ and $\mathbf{R}_n$ are biorthogonal; that is $\mathbf{L}_i^T\mathbf{R}_j = 1$ if $i = j$ and zero otherwise. Where $\lambda_0 = 1 > \lambda_1 \ge \cdots \ge \lambda_{N-1} \ge 0$ are the sorted eigenvalues. The largest eigenvalue, i.e., $\lambda_0 = 1$, has been associated with right eigenvector $\mathbf{R_0} = \mathbf{1}_n$ and left eigenvector $\mathbf{L_0}$. Note that not all eigenvectors are equally important. The eigenvector $\mathbf{R_0} = \mathbf{1}_n$ does not contribute to the computation of diffusion distance, therefore, it can be ignored [36]. Thus, $\mathbf{P}$ is diagonalizable as the product, $\mathbf{P} = \mathbf{R}\mathbf{\Lambda}\mathbf{L}^T$, where $\mathbf{R} = (R_1, \cdots, R_{N-1})$, $\mathbf{L} = (L_1, \cdots, L_{N-1})$, and $\mathbf{\Lambda} = diag\{\lambda_1, \cdots, \lambda_{N-1}\}$. Using the nontrivial eigenvalues and right eigenvectors of $\mathbf{P}$, the diffusion distance can be rewritten as

$$D^{(t)}\left(x_i, x_j\right) = \sqrt{\sum_{n=1}^{N-1}\lambda_n^{2t}\left(\mathbf{R}_n(x_i) - \mathbf{R}_n\left(x_j\right)\right)^2}. \tag{10}$$

The eigenvectors corresponding to larger eigenvalues usually can capture more information from the data. So we can actually use fewer numbers of eigenvectors to span a low dimensional space. As mentioned above, the target dimensionality of the low-dimensionality representation usually is set equal to the intrinsic dimensionality d'. Naturally, an embedding map $\Re_{DM}$ from the node $x \in R^d$ on the graph $\mathbf{G}$ to a vector x' in the Euclidean space $R^{d'}$ $(0 < d' \le d)$ can be constructed as follows:

$$D^{(t)}\left(x_i, x_j\right) \approx \sqrt{\sum_{n=1}^{d'}\lambda_n^{2t}\left(\mathbf{R}_n(x_i) - \mathbf{R}_n\left(x_j\right)\right)^2} = \sqrt{\left\|\Re_t(x_i) - \Re_t\left(x_j\right)\right\|^2}, \tag{11}$$

and

$$\Re_{DM} : x \xrightarrow{\Re_{DM}} x' = \left[\lambda_1^t\mathbf{R}_1(x), \lambda_2^t\mathbf{R}_2(x), \cdots \lambda_{d'}^t\mathbf{R}_{d'}(x)\right]^T. \tag{12}$$

Eq. (11) indicates that the Euclidean distance in the low dimensional space is approximately equal to the diffusion distance in the original

space. In this way, the low dimensional data representation $x' \in R^{d'}$ is obtained, and the dimensionality reduction is accomplished.

3.3. Diffusion maps based k-nearest-neighbor rule fault detection scheme

Unlike FD-kNN which directly uses the original data set, DM-kNN uses correlation dimension method to estimate the intrinsic dimensionality of normal training data set. Then diffusion maps analysis is applied to the high dimensional original data set to extract the low dimensional manifold feature with the intrinsic dimensionality. After diffusion maps analysis, the redundancy of the original data set is removed, thus, the burden of data storage is reduced. As a rule, too much redundant information in the data set may also blur the intrinsic variables and deteriorate the detection performance. That's why FD-kNN has been too highly missed detection rates. Moreover, diffusion maps are a nonlinear technique, it can extract nonlinear relations or features of the severe non-linearity semiconductor data compared with PCA. At the same time, diffusion maps could obtain the low dimensional embedding manifold of the data through reserving diffusion distance. Also diffusion distance is computed using all possible paths through the graph and considering the contribution of all the edges that connect the specified point. These two aspects make the feature extraction much more robust and DM-kNN method much more stable for fault detection. In addition, normal and fault samples that belong to different process conditions have different properties, and they are obviously in different manifold regions after diffusion maps analysis. Coincidentally, the kNN detection statistic is based on distance, which causes DM-kNN to have an optimal fault detection performance. The proposed detection approach consists of two procedures: model building and fault detection. The schematic diagram is shown in Fig. 2, and the detailed step-by-step procedure is summarized below.

Modeling phase:

1. Obtain the three-way normal batch process data set **X**, unfold it through batch-wise, and normalize it.
2. Calculate the intrinsic dimensionality $d'(0 < d' \leq d)$ of the data set **X** using correlation dimension method.
3. Extract the low dimensional manifold feature space $\mathbf{X}' \in R^{d'}$ by diffusion map analysis.
4. Find k-nearest neighbors for each sample in the low dimensional manifold feature space **X**′ and calculate the kNN squared distance D_i^2 based on Eq. (1).

4. Estimate the corresponding control limit of D_i^2 for fault detection

For the determination of threshold, He and Wang [16] assume that the k-nearest neighbor distances d_{ij} are normally distributed around a nonzero mean among training samples and D_i^2 follow a non-central χ^2 distribution. Thus, the threshold with a significance level can be estimated by the chilimit function. However, because of the complexity of the industrial process, the process sample is impossible to obey the rigorous statistical distribution. Also the selection of the k-nearest neighbor distances of each sample is not simple algebraic operation, which may also relate to the material and energy conservation, etc. of the production process. Under this condition, it is difficult to use a specific distribution to calculate the threshold of the distance metric. In this paper, kernel density estimation is used to estimate the corresponding control limit. Kernel density estimation is a non-parametric estimation, which

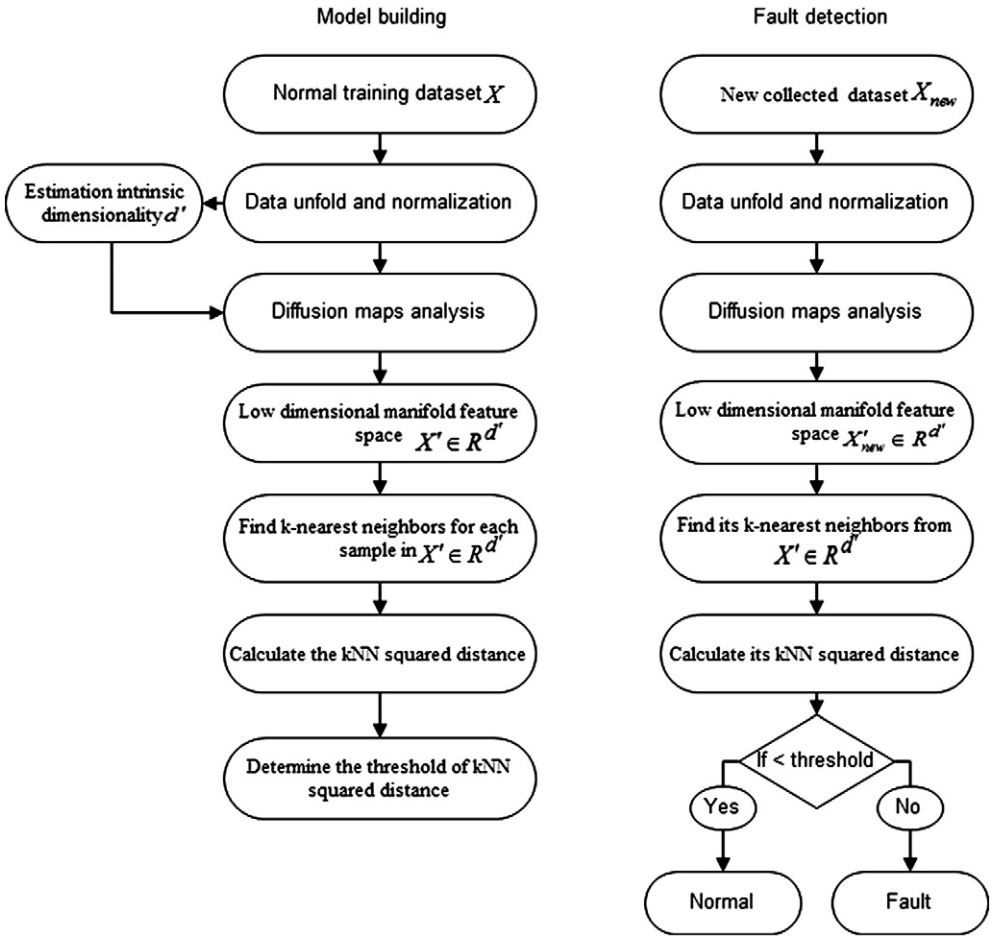


Fig. 2. Flowchart of the proposed diffusion maps based k-nearest-neighbor rule method.

does not require prior knowledge of the data, and does not make any assumptions. In practice, it usually requires only a few samples to estimate the underlying distribution of the data [43]. The kernel density estimator is defined as

$$\hat{f}\left(D_X^2, h\right) = \frac{1}{Nh}\sum_{i=1}^{N} K\left(\frac{D_X^2 - D_i^2}{h}\right), K(x)>0, \text{ and } \int K(x)dx = 1 \tag{13}$$

where h is the bandwidth and K denotes the Gaussian kernel function. The 95% or 99% confidence limit C_α with the significance level $\alpha = 0.01, 0.05$ can be determined by

$$\int_{-\infty}^{C_\alpha} \hat{f}\left(D_X^2, h\right)dx = 1-\alpha. \tag{14}$$

Detection phase:

1. Scale the new collected unclassified data $\mathbf{X}_{new}$ using the mean and variance values obtained from the training data.
2. Extract the d' dimensional manifold feature $\mathbf{X}'_{new}$ by diffusion maps analysis.
3. Identify $\mathbf{X}'_{new}$'s k-nearest neighbors for each sample in the feature space $\mathbf{X}'$ and calculate the kNN squared distance D_x^2 using Eq. (1).
4. Compare D_x^2 with the threshold D_α^2, if $D_x^2 \le D_\alpha^2$, it is considered as a normal sample. Otherwise, it is detected as a fault.

5. Application example

In this section, two process cases are given to illustrate the performance of DM-kNN for process fault detection. First, the simulation examples with nonlinear distribution are used to test the effectiveness and robustness of diffusion maps for dimensionality reduction and information extraction. Then, semiconductor manufacturing process is used to verify the fault detection performance of DM-kNN method, and to compare it with MPCA, FD-kNN, PC-kNN and FS-kNN.

5.1. Information extraction for nonlinear processes

The first simulation example is a nonlinear case of three variables described by the following process mode, originally introduced by [44].

$$\begin{aligned} x_1 &= t + e_1 \\ x_2 &= t^3 - 2.5t^2 - 2t + e_2 \\ x_3 &= -4t^3 + 3t^2 + e_3 \end{aligned} \tag{15}$$

Where e_1, e_2 and e_3 are zero-mean white noises with a standard deviation of 0.01 and $t \in [0.01, 2]$. To test the feasibility of the proposed method, a fault case is designed by a linearity of $0.05(k - 100)$ which is added to x_1 from sample 101 to sample 300, where k is the sample number.

Experiment 1. DM, PCA, ICA, KPCA, LLE and ISOMAP are implemented to compare the effectiveness of dimensionality reduction and feature extraction. Normal training data and fault test data consisting of 300 samples respectively are generated according to Eq. (15). The intrinsic dimensionality d' of data set is rounded to 2 by the correlation dimension estimator. Fig. 3 shows the scatter plots of the first two vectors generated by DM and other different techniques in the normal and fault data set for dimensionality reduction.

In comparison with linear PCA and ICA and nonlinear KPCA, LLE, and ISOMAP, DM shows better performance in information extraction and then separates well the fault and normal examples. This is due to the fact that DM is a nonlinear dimension reduction method which could effectively capture the meaningful low-dimensional features hidden in the high-dimensional process data and preserve the local structure of data at the same time. So this could make the distance between the points which were not in a neighborhood becomes much more relaxed [45].

Experiment 2. DM, PCA, ICA, KPCA, LLE and ISOMAP are implemented to compare the robustness in dealing with data set with strong noise in dimensionality reduction and information extraction. Both normal training data and fault test data are generated as Experiment 1, but

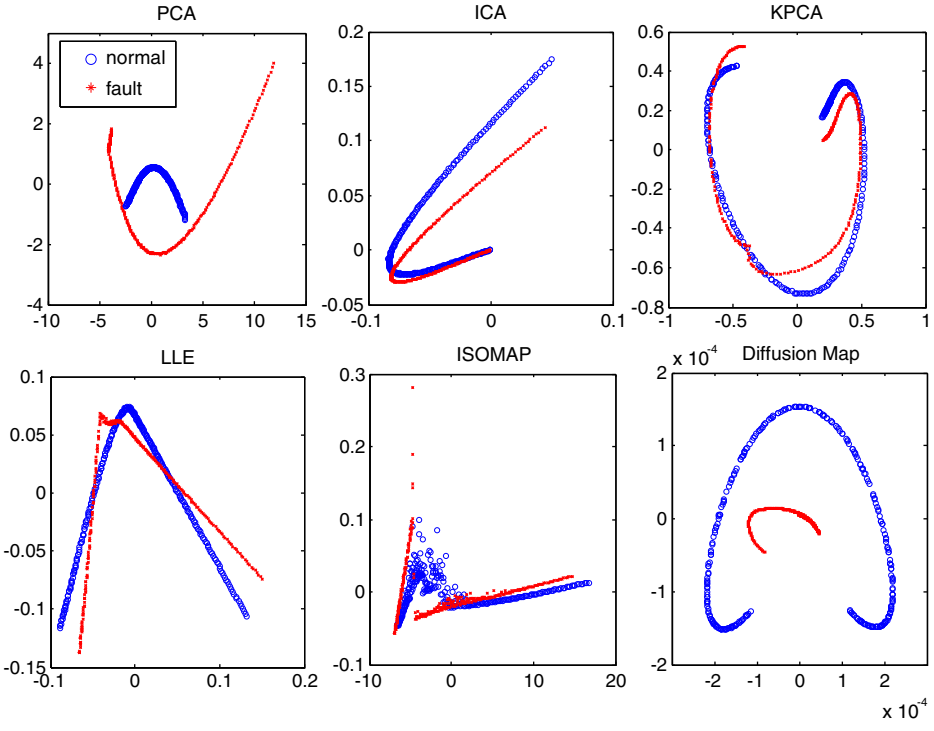


Fig. 3. The dimensionality reduction results of different methods for normal and fault data.

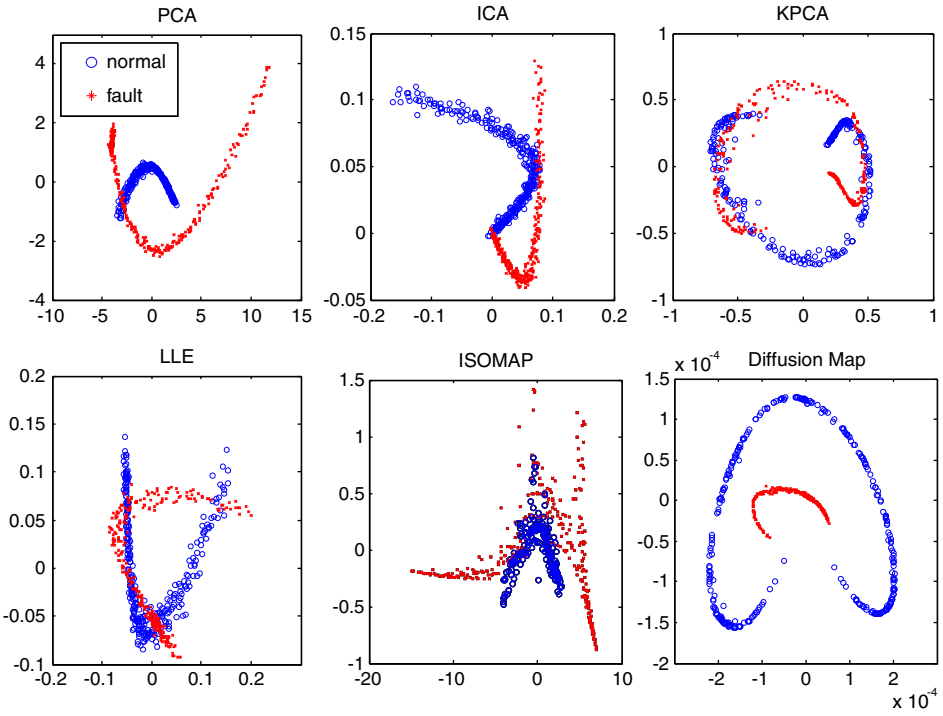


Fig. 4. The dimensionality reduction results of different methods for normal and fault data with strong noise.

the size of the standard deviation of the independent white noise is adjusted to the $N(0,0.1)$. The intrinsic dimensionality d' is also rounded to 2. The scatter plots of the first two vectors generated by DM and other different techniques in the normal training and fault test data set with strong noise for dimensionality reduction are shown in Fig. 4. It can be observed that dimension reduction results of all methods affected by noise appear to have different degrees of deformation in contrast to Fig. 3. However, DM shows minimum deformation and has a higher robustness to small perturbations of the data compared with other methods. This is mainly because DM is based on the diffusion distance which takes into account all possible paths through the graph. In other words, the diffusion distance reflects the connectivity of nodes on the graph and computed based on all the paths between two nodes, so all the "evidence" is considered [33]. Even if one of the paths may be affected by the noise, its impact on the calculation of total diffusion distance is very small due to a relatively little weight ratio. This makes the diffusion distance much more robust to noise than, e.g., the geodesic distance used in ISOMAP, which only considers the shortest path between the two points.

5.2. Semiconductor manufacturing process

In this section, a benchmark semiconductor production process is used to verify the effectiveness of the DM-kNN fault detection method. The process is a typical nonlinear, multimodal complex industrial process. Implementing an effective fault detection technology is very important to reduce the number of scrap and test wafer. In this paper, the data sets are collected from an Al stack etch process performed on a commercial-scale Lam 9600 plasma etch tool at Texas Instrument Inc., which is often used to demonstrate the validity of the batch process detection algorithms [46,47]. Data sets contain 108 wafers of normal data and 21 wafers of fault data respectively and are taken from three different experiments. Fault data sets were intentionally induced through the intervention of the parameters such as RF power, pressure, CI or BCI flow rate during the same experiment environment. Due to a large amount of missing data in two wafers, only 107 normal batches and 20 fault batches are available in this paper. The original data contains 40 variables, including process set points, measured variables, and controlled variables such as gas flow rates, chamber pressure, and RF power [6]. In this paper, 17 process variables which are related to production process and final product status through the analysis of the physics of the semiconductor production process are selected. The process variables used for fault detection are listed in Table 1. The faulty information is listed in Table 2. More detailed description of the semiconductor process and faults can be found in [6].

As discussed in [16], in order to maximize the level of automation and suit industrial applications in fault detection, we use minimum data preprocessing method to obtain equal length batch records in this work. In the first step, the initial 5 sample points are removed to eliminate the possible effect of initial fluctuation in the tool and sensors. Thus, totally 85 sample points are kept to accommodate shorter batches in the data set. This operation is implemented to all training, validation and test data sets. In the following case study, 97 normal batches are randomly selected as the training data for model building. The rest 10 normal batches are used as validation samples and 20 faulty batches are tested for fault detection using the predetermined confidence limits. In this test case, the confidence level is set to 99%. For the comparison, the conventional MPCA, FD-kNN, PC-kNN and FS-kNN methods are also applied to demonstrate the advantage of DM-kNN fault detection

Table 1
Process variables used for fault detection.

No.	Variables	No.	Variables
1	BCl3 Flow	10	RF Pwr
2	Cl2 Flow	11	RF Impedance
3	RF Btm Pwr	12	TCP Tuner
4	Endpt A	13	TCP Phase Err
5	He Press	14	TCP Impedance
6	Pressure	15	TCP Top Pwr
7	RF Tuner	16	TCP Load
8	RF Load	17	Vat Valve
9	RF Phase Err		

Table 2
Induced Faults.

No.	Fault types	No.	Fault types
1	TCP + 50	11	Cl2 + 5
2	RF-12	12	BCl3-5
3	RF + 10	13	Pr + 2
4	Pr + 3	14	TCP-20
5	TCP + 10	15	TCP-15
6	BCl3 + 5	16	Cl2-10
7	Pr-2	17	RF-12
8	Cl2-5	18	BCl3 + 10
9	He Chuck	19	Pr + 1
10	TCP + 30	20	TCP + 20

method. Detailed fault detection results of MPCA, FD-kNN, PC-kNN, FS-kNN and DM-kNN methods are given and analyzed as follows.

First MPCA analysis is carried out on the preprocessed data sets to detect the faults. Totally, three principal components are selected to build the PCA model based on the criterion of cross validation method. The fault detection results based on the T^2 and SPE index are shown in Figs. 5 and 6. For 20 batches of semiconductor fault, we can see that only 3 faults are detected by the T^2 index and 12 faults are detected by SPE index. SPE and T^2 charts together detect 12 faults due to the overlapping phenomenon between the faults detected. Detailed fault detection results are listed in Table 3. It shows that MPCA does not perform well. The main reasons are as follows: one is that the MPCA is a linear approach which is more suitable for linear, unimodal batch process. However, semiconductor process is a severe nonlinear and multimodal distribution process due to process shifts so that the MPCA based detection technique become ill-suited. Another reason that should be noticed is that how well the semiconductor data obey the fundamental assumptions of multivariate Gaussian distribution approximately of the derivation of control limits for the T^2 and SPE detection statistics, which determines the fault detection performance. The T^2 confidence limit is determined upon the assumption that the scores follow a multivariate Gaussian distribution, and the derivation of SPE confidence limit relies on the assumption that the unfolded variables follow a multivariate Gaussian distribution. Nevertheless, the unique characteristics such as multimodal batch trajectories and non-linearity of the semiconductor process make the assumption of multivariate Gaussian distribution for the unfolded variables or the scores violated. Here, the T^2 statistic is used to illustrate this point. The score plot spanned by the first two PCs is shown in Fig. 7. It can be observed that the scores do not obey Gaussian distributions; the "normal" hyper-ellipsoid of 99% confidence

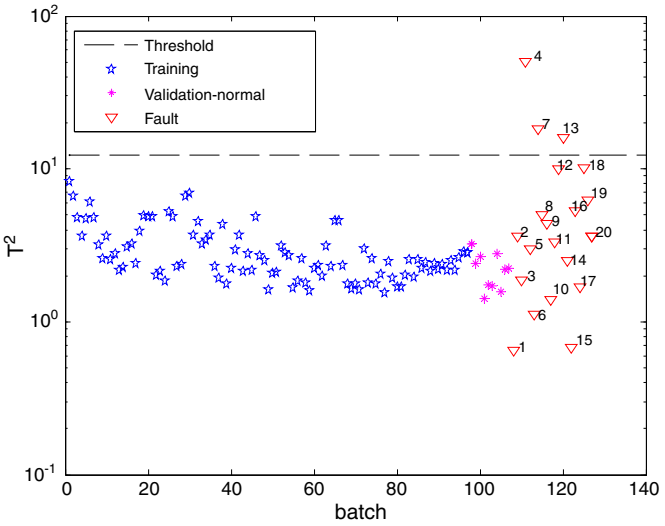


Fig. 5. Fault detection results using MPCA-T^2.

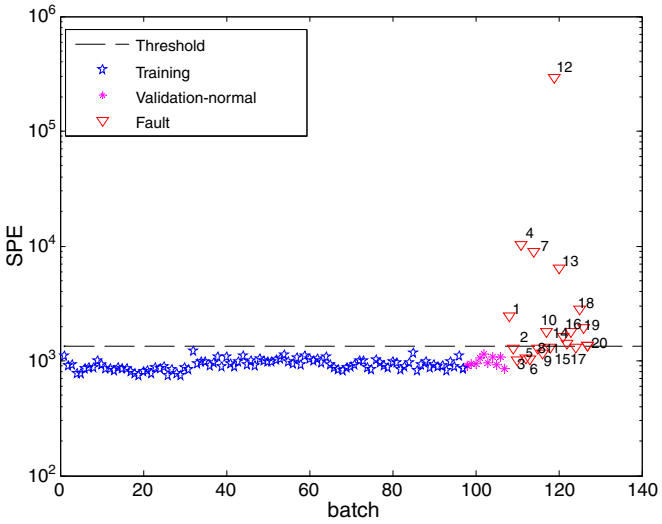


Fig. 6. Fault detection results using MPCA-SPE.

limit calculated by MPCA is expanded and covers part of the abnormal operation region which will lead to elevated Type II error rates. Furthermore, Fig. 7 also shows the non-linearity or multimodal distribution characteristics of the batch trajectories that are grouped into three clusters. The above analysis shows that it is the unique characteristics of semiconductor data that make the MPCA less effective.

Then FD-kNN, PC-kNN and FS-kNN are applied to the semiconductor etch data set. The number of nearest neighbors is set to 3 for all methods, and 3 PCs are used for PC-kNN and FS-kNN based on the criterion of cross validation analysis. The confidence limit of detection statistics D_α^2 is computed by kernel density estimation. In Figs. 8, 9 and 10, the detection results are shown as a semi-log plot. The detailed fault detection results for FD-kNN, PC-kNN and FS-kNN methods are listed in Table 3. It can be seen from Figs. 8–10 or Table 3 that the number of faults detected by FD-kNN, PC-kNN and FS-kNN is 15, 14 and 18. The detection result demonstrates that all the three detection methods based on kNN rules perform better than MPCA. This illustrates that kNN rule based methods outperform MPCA in handling process non-linearity and multimodal distribution that really exist in semiconductor processes. Although FD-kNN can provide a better performance of fault

Table 3
Detection results of different methods for 20 batches of semiconductor faults.

Fault no.	MPCA-T^2	MPCA-SPE	FD-kNN	PC-kNN	FS-kNN	DM-kNN
1	x	✓	✓	✓	✓	✓
2	x	x	x	x	✓	✓
3	x	x	x	x	x	✓
4	✓	✓	✓	✓	✓	✓
5	x	x	x	✓	✓	✓
6	x	x	x	x	✓	✓
7	✓	✓	✓	✓	✓	✓
8	x	x	✓	✓	✓	✓
9	x	x	x	x	✓	✓
10	x	✓	✓	✓	✓	✓
11	x	x	✓	x	x	✓
12	x	✓	✓	✓	✓	✓
13	✓	✓	✓	✓	✓	✓
14	x	✓	✓	x	✓	✓
15	x	✓	✓	✓	✓	✓
16	x	✓	✓	✓	✓	✓
17	x	x	✓	✓	✓	✓
18	x	✓	✓	✓	✓	✓
19	x	✓	✓	✓	✓	✓
20	x	✓	✓	✓	✓	✓
Detection results	*3*	*12*	*15*	*14*	*18*	*20*
Type II error rate	*85%*	*40%*	*25%*	*30%*	*10%*	*0*

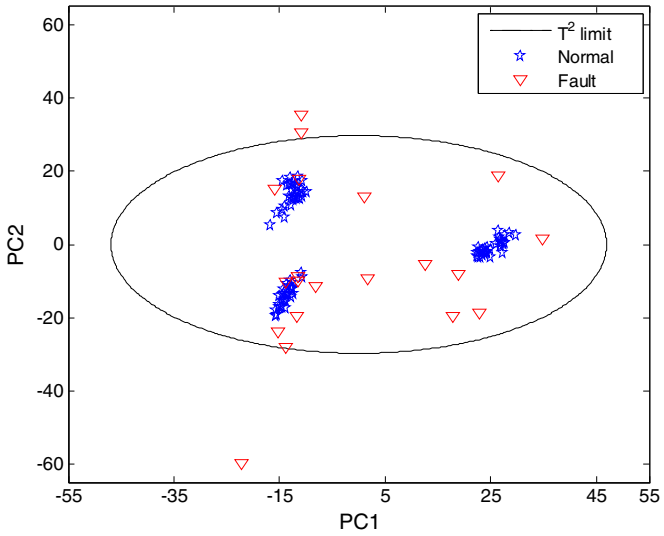


Fig. 7. Scatter plot of the first two principal component of PCA.

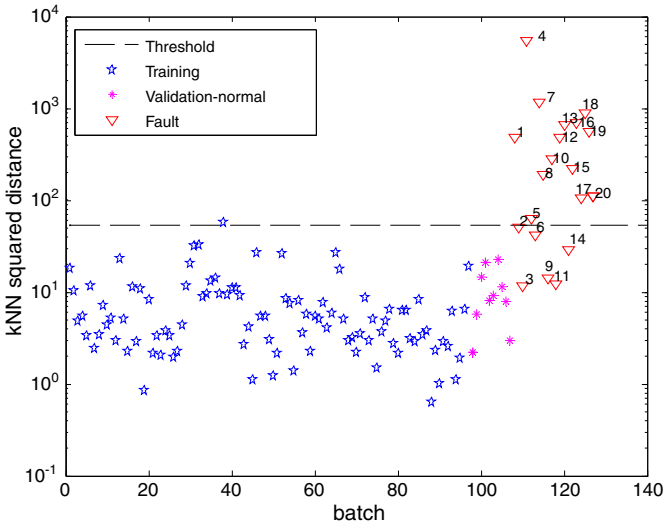


Fig. 9. Fault detection results using PC-kNN.

detection compared with MPCA, it still has a high missed detection rate. This is mainly because FD-kNN is applied directly to the original data sets, which the redundancy in the data sets may also blur the intrinsic variables and deteriorate the detection performance. Moreover, it is worth noting that PC-kNN works similar to the FD-kNN in detecting faults, but the saving in storage of the training samples to compute D_x^2 for each test sample is significantly improved. However, PC-kNN has a higher Type II error rate when compared against the performance of FS-kNN. Because PC-kNN makes no use of residual subspace where any fault occurs will not be detected. In contrast, FS-kNN fully takes into account this residual information, which reveals important process properties for fault detection. This is the reason that FS-kNN outperforms PC-kNN. Nevertheless, both PC-kNN and FS-kNN cannot detect all 20 faults and have a high Type II error rate. The major factor that affected the performance of the PC-kNN and FS-kNN fault detection methods is that the above PCA based information extracting strategies only consider the linear features of process data and do not effectively capture the nonlinear characteristics in semiconductor industrial processes.

Finally, the proposed DM-kNN is applied to the same data set, and the number of nearest neighbors is also set to 3. The intrinsic dimensionality of the preprocessing data set is rounded to 18 by the correlation dimension estimator. Fig. 11 shows the scatter plots of the normal and fault batches spanned by the first two vectors generated by DM technique for dimensionality reduction. Fig. 11 shows that normal batches are grouped in one cluster and fault batches evidently dissociate away from the cluster of normal batches. Obviously, as a result of normal and fault samples that belong to different process conditions and have different properties, they should be located in different areas of the manifold in low feature space after diffusion maps. In other words, the fault features of fault samples are amplified when diffusion maps analysis is applied. The phenomenon is presented as the fact that the distance between the fault samples and the normal samples in the feature space is further. Since kNN is based on distance measurement technology, the farther the distance between the fault and normal samples, the more likely, it is to detect the faults. The detection result is shown in Fig. 12, and it shows that the proposed method successfully detects all 20 faults, which all of them are far from the control limit of 99%. Compared to detection results of the MPCA, FD-kNN, PC-kNN and FS-kNN methods, the detection performance has been further improved by DM-kNN. In addition, Table 3 presents the Type II error rates of all the detection methods. It can be concluded that DM-kNN has the smallest Type II error rate and provides superior fault detection performance.

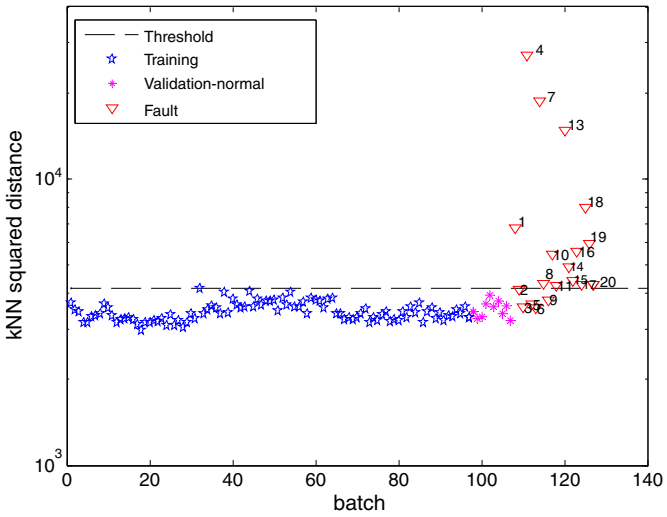


Fig. 8. Fault detection results using FD-kNN.

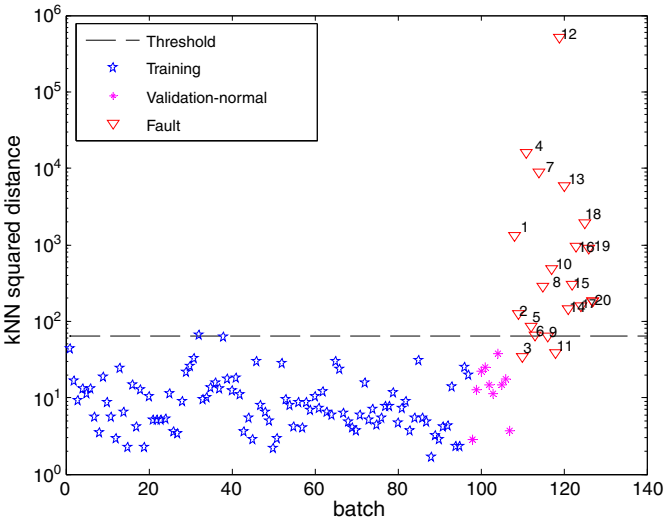


Fig. 10. Fault detection results using FS-kNN.

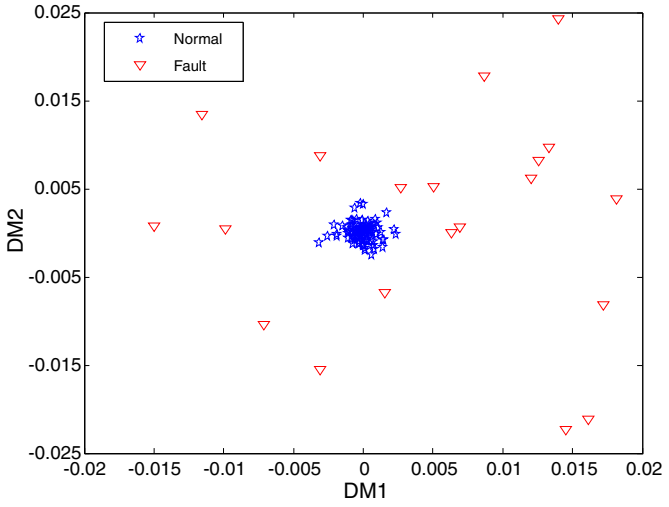


Fig. 11. Scatter plot of two vectors generated by diffusion maps.

Now, it is interesting to discuss the main reasons that lead to the different fault detection performances of PC-kNN, FS-kNN and DM-kNN. Note that they are all based on the unified framework of kNN, the main difference lies in the way of feature extraction or dimensionality reduction. Compared with PCA, DM exhibits two major advantages in the feature extraction process: it is nonlinear, and it preserves local structures. In most cases, the first aspect is important since the original forms of the raw observations are severely nonlinear, and the linear methods (such as PCA) could not discover the nonlinear degrees of freedom that underlie the given data. Moreover, on the basis of the fact that in many applications, distances of these feature points that are far apart do not contribute meaningfully to data analysis, so preserving the intrinsic local geometric structures is needed in the feature extraction process [31]. Conversely, PCA for dimensionality reduction through eigenvalue decomposition of the covariance matrix, where the distances of all possible pairs of points are considered, and it is likely to retain the larger distance in the feature space. So PCA is a global technique which does not preserve the local distances in the feature extraction process [48]. Unlike PCA, DM for dimensionality reduction via spectral eigenvalue decomposition of the Markov transition matrix directly reflects the local geometry structures of data set. The Markov transition matrix is constructed by defining a Markov chain on the graph of the data set using the weighted matrix (also weight function is specified by kernel function). The weight matrix penalizes pairs of points that are far away from each other, and forces pairs of adjacent points to be mapped very close. Additionally, the Markov chain on the graph can be viewed as the propagation of the local influence of each node with its neighbors. So the local geometric structures can be preserved in such a nonlinear way in a diffusion process. Conclusively, the eigenvectors of the Markov transition matrix are much more meaningful in the analysis of data than PCA as they relate to the diffusion and inference on the data. Hence, selecting an appropriate feature extraction technique is often necessary before the implementation of fault detection statistics, which makes an important impact on the final detection results.

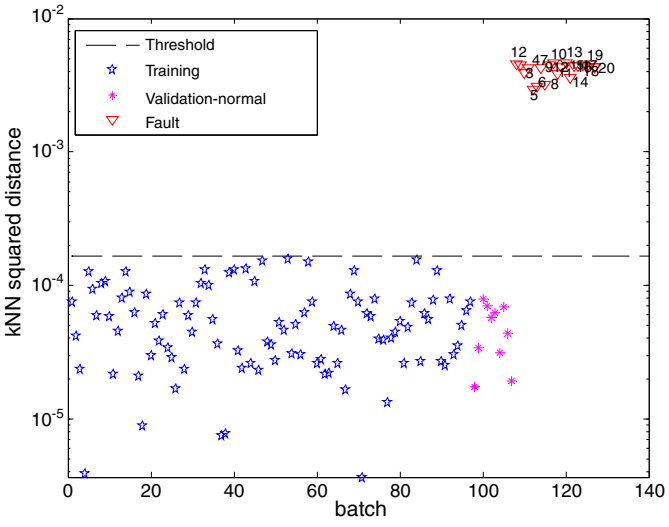


Fig. 12. Fault detection results using DM-kNN.

6. Conclusion

In this paper, a novel diffusion maps based k-nearest-neighbor rule (DM-kNN) technique is proposed and applied to the fault detection of semiconductor manufacturing processes. The presented approach uses the correlation dimension estimator to assess the intrinsic dimensionality of the original process data. Then, the diffusion maps can help extract the low dimensional manifold features with the intrinsic dimensionality that optimally preserves the intrinsic nonlinear structure of the data sets. Subsequently, the adapted kNN rule is applied to the low dimensional manifold feature space to perform fault detection.

The DM method is applied to a nonlinear simulation example for dimensionality reduction and information extraction. The dimensionality reduction results demonstrate that the DM algorithm is superior to PCA, ICA, KPCA, LLE, and ISOMAP methods. As for coping with system noise, the DM shows the most reliable and robust information extraction capability. The presented approach is applied to the semiconductor manufacturing processes, and the detection results are compared against those of MPCA, FD-kNN, PC-kNN and FS-kNN methods. The application results demonstrate that the developed DM-kNN approach can effectively monitor the complex semiconductor manufacturing processes with superior fault detection performance. The next of our future work is to explore the fault diagnosis capability of the DM-kNN method after process faults are detected. In addition, the further application of the proposed approach in other chemical processes is still underway.

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